INCH-POUND

MIL-P-70551 (AR)
AMENDMENT 2
16 November 1999
SUPERSEDING
AMENDMENT 1
31 July 1993

MILITARY SPECIFICATION

FOR
PROJECTILE, 81MM SMOKE, RP,
M819 METAL PARTS

MIL-P-70551(AR) was inactivated after 04 January 1999 for new design.

This Amendment forms a part of MIL-P-70551 (AR), dated 5 September 1986, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1.1, Under "SPECIFICATIONS", Add:

"FEDERAL
TT-C-490 - Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings."

PAGE 3

* 2.2, Under "ASTM Method, add:

"B117 - Standard Test Method of Salt Spray (Fog) Testing."

PAGE 5

4.3.1, Delete: "Shear Pin, 9300226, 10".

* Delete: "... 2 support disc"

AMSMC N/A FSC 1315

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:40Z
Check the source to verify that this is the current version before use.

* Delete: "4" under "Quantity" column, and substitute: "2".

* Add:

"Round-type tensile test specimens from support disc bar stock ...(-) ...2".

PAGE 6

TABLE I: Delete all shear pin callouts in their entirety.

* Add:

"Body Tube Assembly (dwg. 9327832)
Examination for defects ...5 ...100% ...3.2 ...4.4.2.6
Paint adhesion and salt spray test ...2 ...100% ...3.2 ...4.5.4".

PAGE 7

4.4.1.a, Delete in its entirety.

PAGE 8

* 4.4.2.1,

Major 101, No. of Sample Units, Delete: "2 1/", and substitute: "1/"

Major 102, No. of Sample Units, Delete: "2 1/", and substitute: "2/"

Major 103, Delete in its entirety and substitute:

"103 Tension test (rear support disc) ...3/ ...100% ...3.1 ...4.5.2.3".

Note 1/, Delete in its entirety and substitute:

"1/ See 4.5.2.1."

Add:

"Note 2/ See 4.5.2.2."

"Note 3/ See 4.5.2.3."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:40Z
Check the source to verify that this is the current version before use.

PAGE 10

* 4.4.2.2, Major 123, after "groove", add "at basic depth".

Add:

"Major 126 Presence of burrs or foreign matter in interlock channel ...(-) ...0.40% ...3.2 ...Visual".

"Major 127 Angle at rear of obturator groove ...(-) ...0.40% ...3.2 ...Gage".

PAGE 12

4.4.2.3, Delete in its entirety.

PAGE 16

* 4.4.2.6, Major 103, Delete examination callout in its entirety and substitute:

"Runout of thread P.D. at projected length with read I.D. and rear and front faces".

Add:

"Major 112 Silastic missing or inadequate from adapter/body tube joint ...2/ ...(-) ...3.2 ...Visual".

"113 Paint adhesion/salt spray test ...3/ ...(-) ...3.2 ...Visual/4.5.4".

"Note 2/ Section 2 units per lot and at first article."

"Note 3/ See 4.5.4."

PAGE 17

* 4.4.2.7, Add:

"Major 106 Presence of lubricant or cleaning fluid on plate ...(-) ...0.40% ...3.2 ...Visual/Manual".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:40Z
Check the source to verify that this is the current version before use.

* 4.4.2.8, Major 112, No. of Sample Units,

Delete: "2", and substitute: "1/"

Add: "Note 1/ See 4.5.3.2."

* 4.5.2.2, Delete text after "Tail cone" in its entirety and substitute:

"From each of two tail cones selected from one interfix, one tension test specimen, the longest flat or round type obtainable, shall be taken longitudinally from the tapered portion of the component. Specimens, equipment and procedures shall be in accordance with ASTM E-8. Failure of any specimen to meet its tensile strength or elongation requirement shall cause the entire applicable quantity of material to be rejected. If a machining defect is suspected as the cause of failure, another specimen may be substituted."

* 4.5.2.3, Delete text after "Support disc" in its entirety and substitute:

"One standard size round-type tensile specimen shall be taken longitudinally from the center of the bar's diameter from each 1000 feet of material or fraction thereof. For first article, two specimens shall be tested. Specimens, equipment and test procedures shall be in accordance with ASTM E-8. Rejection and retest criteria shall be as specified in 4.5.2.2."

* 4.5.3, Delete in its entirety.

* Add new paragraph 4.5.4 as follows:

"4.5.4 Paint adhesion/salt spray tests. In performing the following tests, TT-C-490 shall apply unless specified otherwise herein. The tests shall be performed during first article inspection, each day of painting during production and whenever a change is made to the paint process, until all samples from five (5) consecutive days of painting have passed the tests. Thereafter, the sample frequency may drop to twice per week, during continuous days of painting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:40Z
Check the source to verify that this is the current version before use.

The specimens shall consist of painted body tube assemblies, randomly selected from assemblies painted at the same time and in the same equipment as used in regular production. The samples shall be taken from each continuous final paint operation up to 24 hours long. One projectile shall be taken during the first 30 minutes of painting, and the other at any other time.

The paint adhesion test shall be performed on each projectile prior to salt spray testing using the procedures and pass/fail criteria contained in 4.2.8 and 3.5.6, respectively, in TT-C-490. After passing the adhesion test, each specimen shall be scored lengthwise using a precision scribing tool/insert or equivalent in preparation for the salt spray test.

The duration of the salt spray test shall be as specified on drawing 9327832. The salt solution, test equipment, and drying procedures shall be in accordance with ASTM B117, Salt Spray (Fog) Testing, unless specified otherwise herein. Specimens shall be positioned between 15 and 30 degrees from vertical during the test. No coating of any kind shall be permitted over the paint on stamp marks. However, areas in contact with racks or supports, unpainted surfaces and areas within 1/8 inch of unpainted surfaces may be protected. The significant surface shall be as specified in 4.2.9 of TT-C-490. Although the failure criteria in a-g below applies to the significant surface, the government may also reject a specimen for variations of sections a-g, or for an unlisted defect(s) anywhere on the part, if deemed excessive.

As used herein, the term "blister" refers to an unbroken area of paint with a bubble-like appearance, whereas "spots" are exposed phosphate pretreatment, bare metal, or corrosion (rust) spots breaking through the paint. Blisters which are present when the test is over are to be counted within five (5) minutes of removal from the test chamber. After testing, any projectile which fails for any of the following reasons shall be cause for rejection of all parts painted since the last successful test, and a return to the original sampling plan if failure occurred during the reduced sampling phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:40Z
Check the source to verify that this is the current version before use.

a. The paint shows more than 1/8 inch creepage, blistering, or loss of adhesion of paint from the scribe mark, stamp mark, or edge.

b. There are more than two (2) scattered blisters.

c. Any blister exceeds 3/64 inch average diameter.

d. Any spot exceeds 3/32 inch average diameter.

e. There are three (3) or more scattered spots exceeding 1/16 inch average diameter.

f. There are five (5) or more spots, total.

g. There is evidence of rust under the paint (a brown discoloration showing through the paint) in excess of 1/4 inch in diameter total area.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, deletions, modifications, corrections) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-0013)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:40Z
Check the source to verify that this is the current version before use.